

CO2-AT3 Class Test

Pool A

2 marks Questions

75

① Primary key:

primary key is a column (or a combination of columns) that uniquely identifies each row in a database table.

Properties:

1. Uniqueness
2. Entity Integrity
3. Data Referencing

② CRUD operations

CRUD Operation	SQL Equivalent	Function
Create	Insert	Adds new records
Read	Select	Retrieves existing records.
Update	Update	Modifies existing data
Delete	Delete	Removes specific data

③ SQLite: Serverless, file-based embedded DB,
Zero setup; best for local mobile apps.

MySQL: Server-client relational database; high performance; high handles concurrency.

④ Foreign key definition & example.

Def: A column in a child table that references the primary key of a parent table to maintain referential integrity.

ex:

SQL
CREATE TABLE Orders (

OrderID INT Primary KEY,

CustID INT,

FOREIGN KEY (CustID) Ref Cust (CustomerID)

⑤ Importance of Database Normalization

Eliminates Redundancy: Reduces duplicated storage across table.

Prevents Anomalies: Protects against Insertion, Update, Delete anomalies during operations.

Part B - 10 Marks Questions

⑥ University Database Analysis

Relationship Type: One-to-Many (1:M)

One department contains many students.

Each student belongs to one department.

Primary & Foreign Key:

* PK: Regardless each student & department has a unique, non-null identifier.

* FK: Links students to the Department table.

Data Integrity:

Referential Integrity: Prevents assigning a student to a department ID.

Cascade Constraints: Restricts deleting a department if active students are still associated with it.